3 **(a)** $\omega = \frac{v}{r} = \frac{0.240}{0.080} = 3.0 \text{ rad s}^{-1}$

(b) **(i)** Consider vertical equilibrium of the bob,

$$T\cos\theta = mg$$

$$T = \frac{mg}{\cos\theta}$$

$$= \frac{1.5\times 9.81}{\cos 30°}$$

$$= 16.991 = 17.0 \text{ N}$$

(ii) The horizontal component of tension provides the centripetal force for the bob's horizontal circular motion.

$$T\sin\theta = m\omega^2 R$$

$$T\sin\theta = m\omega^2\left(d + L\sin\theta\right)$$

$$d = \left(\frac{T}{m\omega^2} - L\right)\sin\theta$$

$$= \left(\frac{16.991}{1.5\times 3.0^2} - 0.250\right)\sin 30°$$

$$= 0.5043 = 0.504 \text{ m}$$

(c) The weight of the bob is always acting vertically downwards. Hence the tension in the string needs to have an upward vertical component to keep it in vertical equilibrium. So the string cannot be horizontal and θ must be smaller than $90°$.

The statement is valid.

Raffles Institution
Year 5-6 Physics Department

4 **(a)** The gravitational force due to a mass is always attractive. Hence, the external force